TRACKING U.S. SUPPLY CHAIN CONGESTION

GS Supply Chain Congestion Scale: July 6th; Index Higher W/W, Bottleneck Scale Unchanged at ‘2’

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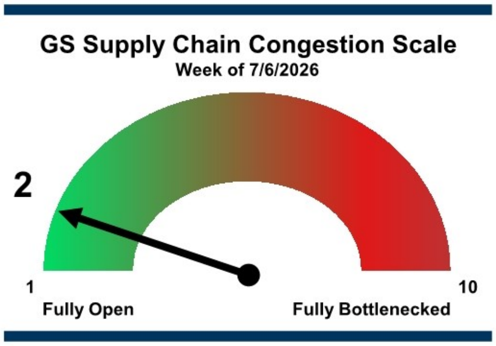
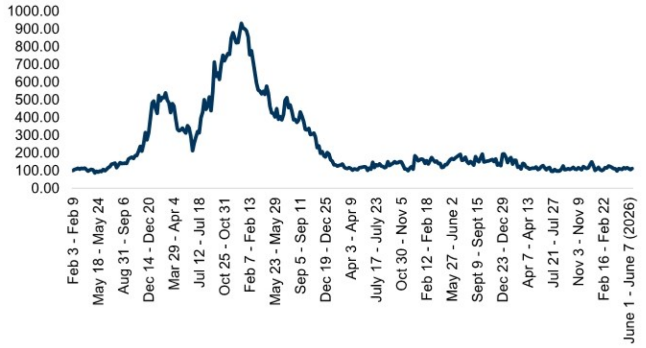


Scale is based solely off weekly metrics to give more granularity on high frequency data indications; see Appendix for scale that combines weekly and monthly metrics

Source: Goldman Sachs Global Investment Research

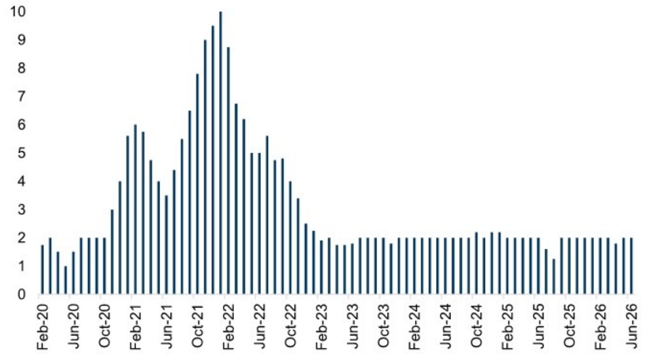
Our weekly bottleneck scale remained at ‘2’ this week, though the absolute level of our congestion index increased moderately on a sequential basis (+8% w/w; Exhibit 2). For this week’s scale and index, the number of container ships waiting to dock and unload goods along the West Coast rose from 1 to 2, while backlogs along the East Coast declined from 4 to 3 (Exhibit 6). West Coast rail intermodal traffic growth decelerated versus last week (+10% YoY vs +14% YoY last week; Exhibit 7), while rail service metrics were mixed. Chassis dwell times decreased slightly on average at US ports (Exhibit 9), while ocean container shipping rates (China to US West Coast) were up +8% WoW and up +82% YoY (Exhibit 10).

Exhibit 1: Our weekly composite index increased in the most recent week (+7% w/w); the bottleneck scale remained at ‘2’; overall bottleneck levels remain well below peak congestion levels when scale was at ‘10’ and now imply levels in line with pre-Covid fluidity
GS Weekly Bottleneck Index, Feb 2020 - June 2026



Source: Goldman Sachs Global Investment Research

Exhibit 2: Our average weekly bottleneck score in June is tracking at 2.0; this result is down significantly from the Dec21/Jan22 peak of congestion and close to in line with the pre-Covid baseline
GS Weekly Congestion Scale, Scored by Month*



*Numbers reflect the average weekly score seen in each respective month

Source: Goldman Sachs Global Investment Research

As a reminder (and to help reiterate why and how we construct the index), please refer to the Appendix following Exhibit 17. Additionally, for further clarity on tracked congestion metrics, please refer to the glossary following Exhibit 19.

Transport Subsectors to Watch as Congestion Remains Muted

The key question remains as to what impact tariffs and geopolitical conflicts will have on demand and timing of freight flows, as well as ability to normalize around global trade. Please see our recent tariff impact tracker for additional commentary. Should supply chain pressures broadly continue to mitigate, then it is conceivable we could see the index remaining more consistently in ‘1’ territory in 2026.

Indicator Updates

Of the metrics we track (Exhibit 3), we provide updates for the weekly and monthly variables below (Exhibit 4 - Exhibit 5).

Exhibit 3: Tracked Congestion Metrics

Tracked Metrics	
Weekly Variables	**Monthly Variables**
Container Ships Waiting to Dock at LA/LB	Container Weighted Average Dwell (San Pedro's Bay)
Container Ships Waiting Along to U.S. Gulf/East Coast	Containers Dwelling > 5 days (San Pedro's Bay)
UNP Intermodal Traffic	PMI Manufacturing Supplier Delivery Time
UNP Intermodal Velocity	Big Three' West Coast Ports' Inbound Loaded Containers
UNP Intermodal Dwell	LMI Transportation Capacity
BNSF Intermodal Traffic	LMI Warehousing Capacity
BNSF Intermodal Velocity	LMI Warehousing Utilization
BNSF Intermodal Dwell	Door to Door Shipping Days, China to US
Ocean Shipping Rates, East Asia to U.S. West Coast	Class 8 Trucking Driver Count Growth
Chassis Street Dwell, 20ft Containers	Rail Container Dwell (San Pedro's Bay)
Chassis Street Dwell, 40/45ft Containers	
Chassis Terminal Dwell Time, 20ft Containers	
Chassis Terminal Dwell Time, 40/45ft Containers	

Source: Goldman Sachs Global Investment Research

Exhibit 4: Bottleneck metrics indicated mixed results this week versus last

	Mar 23 - Mar 29	Mar 30 - Apr 5	Apr 6 - Apr 12	Apr 13 - Apr 19	Apr 20 - Apr 26	Apr 27 - May 3	May 4 - May 10	May 11 - May 17	May 18 - May 24	May 25 - May 31	Jun 1 - Jun 7	Jun 8 - Jun 14	Jun 15 - Jun 21	Jun 22 - Jun 28
Container Ships Waiting to Dock at LA/LB	1	0	1	0	1	1	0	1	1	1	1	1	1	2
Container Ship Backup (East and Gulf Coast)*	5	4	4	1	2	2	4	5	4	5	5	3	4	3
UNP Intermodal Traffic (YoY Growth)	-8.7%	-9.0%	-8.2%	-4.3%	-2.0%	-0.3%	-0.8%	8.4%	8.4%	8.3%	15.6%	10.4%	12.5%	8.9%
UNP Intermodal Velocity (YoY Growth)	8.5%	5.0%	8.3%	7.4%	8.3%	3.9%	2.5%	4.2%	2.9%	0.0%	-1.9%	-2.2%	-1.6%	-2.6%
UNP System Dwell (Hours)	20.3	20.1	19.9	19.6	19.4	20.0	20	19.9	19.5	19.8	19.7	19.4	19.9	19.7
BNSF Intermodal Traffic (YoY Growth)	1.9%	2.6%	2.3%	5.7%	4.7%	7.7%	7.8%	9.7%	18.7%	11.5%	15.9%	11.8%	14.9%	11.3%
BNSF Intermodal Velocity (YoY Growth)	3.3%	-3.1%	-1.6%	-2.4%	-2.1%	-3.9%	-0.6%	-3.7%	6.9%	-8.6%	-10.3%	-7.2%	-6.1%	-5.3%
BNSF System Dwell (Hours)	22.6	22.5	22.1	22.7	22.3	22.1	22.3	23.0	22.8	22.6	22.2	22.1	22.4	22.7
Ocean Shipping Rates, East Asia to U.S. West Coast (YoY Growth)	-0.1%	7.7%	0.9%	13.2%	14.9%	17.4%	18.1%	14.3%	14.1%	15.6%	(11.9%)	(19.3%)	2.7%	82.2%
Chassis Street Dwell, 20ft Containers (Days)	5.3	4.9	4.4	4.2	5.4	4.5	4.2	4.9	4.4	4.7	4.7	4.4	5.1	4.3
Chassis Street Dwell, 40/45ft Containers (Days)	6.5	6.9	5.7	5.8	6.0	5.5	5.7	5.6	5.5	5.8	6.2	6.3	6.5	5.9
Chassis Terminal Dwell Time, 20ft Containers (Days)	10.6	10.6	9.4	9.1	11.4	10.3	10.2	11.5	12.0	12.4	10.8	10	11.1	10.3
Chassis Terminal Dwell Time, 40/45ft Containers (Days)	7.2	7.7	6.3	6.1	6.5	6.2	6.2	6.3	5.5	5.5	5.2	4.9	5.3	4.7

As of 3/28/22, we added and backdated our estimates for the East and Gulf Coast container ship backlog.

Source: Marine Exchange of Southern California, AAR, STB, Freightos, Pool of Pools, Pacific Merchant Shipping Association, Port of Long Beach, Port of Oakland, Port of Los Angeles, LMI, US Bureau of Labor Statistics, IHS Markit, Refinitiv Eikon, compiled by Goldman Sachs Global Investment Research

Exhibit 5: Lagged monthly bottleneck metrics for May indicated mixed results vs April

Monthly Variables	Jan-25	Feb-25	Mar-25	Apr-25	May-25	Jun-25	Jul-25	Aug-25	Sep-25	Oct-25	Nov-25	Dec-25	Jan-26	Feb-26	Mar-26	Apr-26	May-26
Container Weighted Average Dwell (Days)	3.3	2.8	2.8	2.8	3.0	2.6	2.9	2.7	2.8	2.7	2.6	2.5	2.8	2.6	2.6	2.6	2.6
% of Containers Dwelling > 5 days	8%	8%	8%	8%	8%	8%	8%	8%	8%	8%	8%	8%	8%	8%	8%	8%	8%
Rail Container Dwell (Days)	7.1	8.0	6.8	4.7	4.7	3.3	5.2	5.0	4.0	3.3	3.7	5.1	6.1	5.1	4.4	5.1	5.2
Big Three Inbound Loaded Containers (YoY)	23.6%	5.8%	11.6%	9.5%	-10.0%	-4.6%	8.6%	-2.1%	-7.3%	-11.5%	-9.5%	-7.1%	-11.6%	0.6%	-2.0%	-1.0%	29.6%
LMI Transportation Capacity	52.6	55.1	53.6	55.2	54.7	52.4	52.6	57.3	55.1	54.5	50.0	36.9	47.1	41.0	39.2	28.4	31.7
LMI Warehousing Capacity	51.7	50.5	52.3	55.4	50.0	47.8	51.1	50.5	51.6	52.0	54.8	61.2	50.0	50.0	46.0	45.5	50.5
LMI Warehousing Utiliziation	68.3	65.5	59.7	60.1	62.5	62.2	59.4	62.1	65.3	56.5	47.5	42.9	54.4	60.3	59.8	64.4	62.9
Door to Door Shipping Days, China to US	50	51	54	49	48	50	46	44	46	47	47	47	47	47	47	47	47
Class 8 Driver Count	1493.1	1487.6	1491.4	1490.6	1487.7	1482.7	1482.5	1480.2	1472.3	1472.6	1467.8	1467.2	1465.6	1465.1	1464.3	1469.2	1464.8
PMI Manufacturing Supplier Delivery Time (YoY)	0.8	11.6	6.1	6.2	9.8	-0.3	-4.7	2.3	10.8	-0.5	0.1	5.5	-0.4	4.6	6.3	10.7	8.4

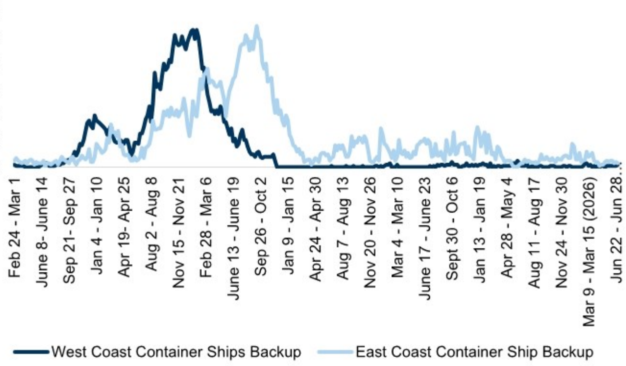
Source: Marine Exchange of Southern California, AAR, STB, Freightos, Pool of Pools, Pacific Merchant Shipping Association, Port of Long Beach, Port of Oakland, Port of Los Angeles, LMI, US Bureau of Labor Statistics, IHS Markit, compiled by Goldman Sachs Global Investment Research

Weekly Indicator Update

Anchored Container Ships

- **West Coast container ship backlogs increased from 1 to 2 in the most recent week while East Coast backlogs declined from 4 to 3.**

Exhibit 6: 3/2* container ships backed up this week on the East/West Coast
West vs. East Coast Container Ship Backlog, Weekly Average, Feb 2020 – June 2026



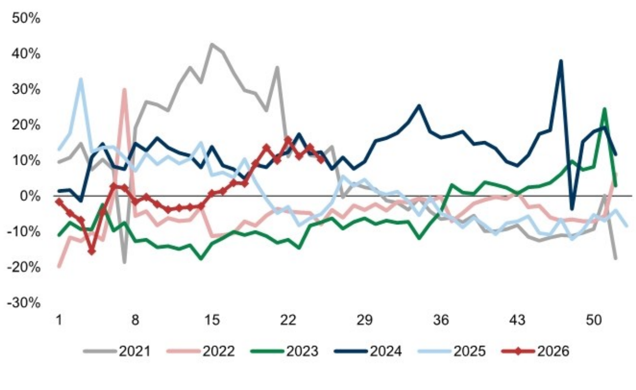
*East Coast is estimated via satellite data – includes container ships sitting for more than 3 days within 140 miles of US ports to the right of longitude 100 (i.e., Gulf and East Coast)

Source: Marine Exchange of Southern California, Refinitiv Eikon, Goldman Sachs Global Investment Research

Rail Intermodal Trends

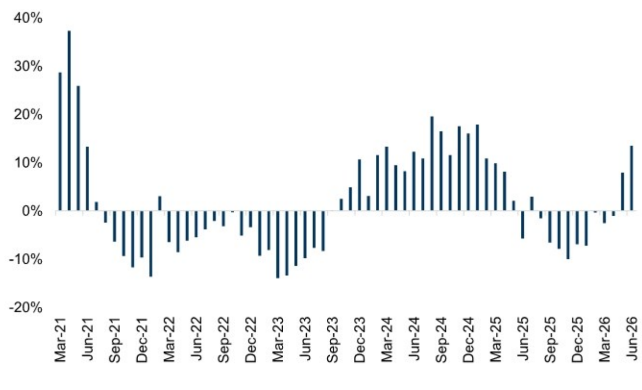
- **West Coast Class 1 Rails' (Union Pacific and Burlington Northern Santa Fe) average intermodal traffic growth decelerated to +10% last week versus +10% in the prior week.**
  - BNSF intermodal traffic at +11% YoY this week vs. +15% YoY last week; UNP intermodal traffic at +8.9% YoY vs. +13% YoY last week.
  - See average BNSF/UNP growth trends in Exhibit 7.

Exhibit 7: West Coast Class 1 Rails' Intermodal Volume Growth, Week 1 – Week 52
YoY % growth



Source: AAR, Data compiled by Goldman Sachs Global Investment Research

Exhibit 8: West Coast intermodal carload growth (UNP and BNSF) is tracking up ~13% YoY on average in June
West Coast Class 1 Rail Intermodal Traffic YoY % Growth



Source: AAR, Data compiled by Goldman Sachs Global Investment Research

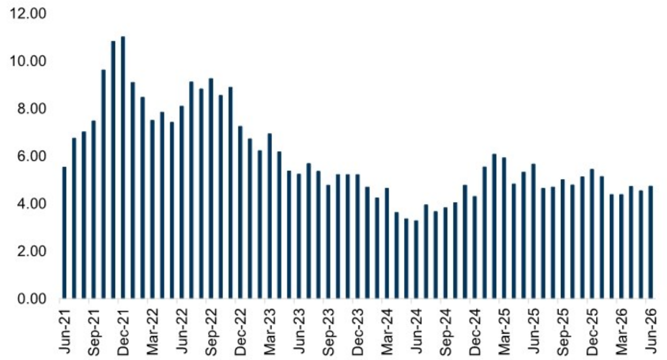
- **West Coast Class 1 Rail terminal dwell and intermodal train speed were mixed in the recent week (detailed below).**
  - UNP terminal dwell decreased from 19.9 hours to 19.7 hours in the most recent week; BNSF dwell increased from 22.4 hours to 22.7 hours.

- On intermodal train speed, BNSF was at -5.3% YoY vs. -6.1% YoY last week; UNP was at -2.6% YoY vs. -1.6% last week.

Chassis Dwell Time

- **Chassis dwell is down significantly when comparing June to peak congestion levels – as per Exhibit 9; dwell data for the most recent week implied improved results vs last week.**
  - Average Street dwell time for 20ft shipping container chassis was at 4.3 days in week 26 of 2026, lower versus 5.1 days in the prior week.
  - Street dwell time for 40/45ft container chassis was 5.9 days in week 26, down versus 6.5 days in the prior week.
  - Chassis terminal dwell time for 20/40ft containers was at 10.3/4.7 days in week 26 versus 11.1/5.3 days in the prior week.

Exhibit 9: Dwell for the more typical 20ft container chassis is well off peak congestion levels
Chassis Street Dwell Time (20ft Containers)



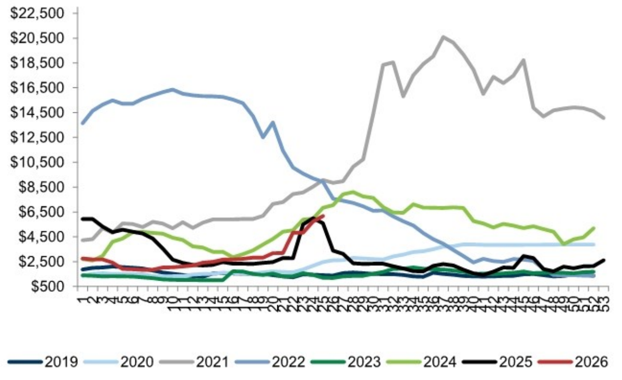
Dwell time shown in days

Source: Pool of Pools

Ocean Shipping Rates

- **The prior week saw ocean container rates at ~$6.18k (+82% YoY), up versus ~$5.74k in the previous week (+2.7% YoY).**

Exhibit 10: Ocean Container Shipping Rates, China/East Asia to North America West Coast



Rate is $ per FEU (Forty-Foot Equivalent Unit)

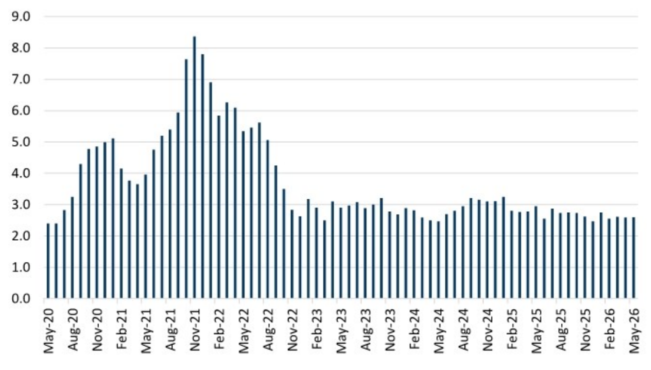
Source: FBX

Lagged Monthly Indicators (May Data)

San Pedro's Bay Container Dwell

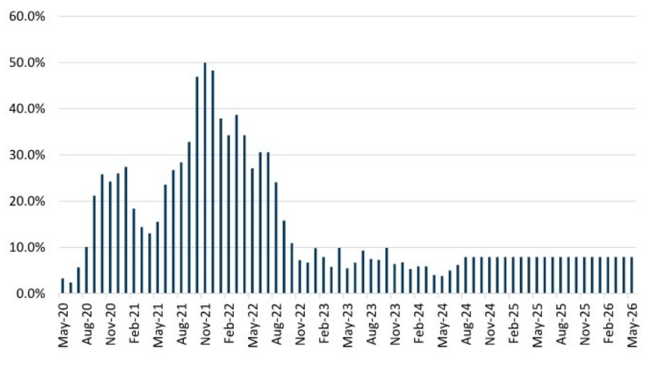
- **Container weighted average dwell time was ~2.6 days in May, unchanged versus ~2.6 days in April.**
- **Rail container dwell was higher in May at 5.2 days vs 5.1 days in April; this result for dwell remains far below 2022's peak rail container dwell of ~16 days.**

Exhibit 11: Container Weighted Average Dwell Time at San Pedro's Bay, Days



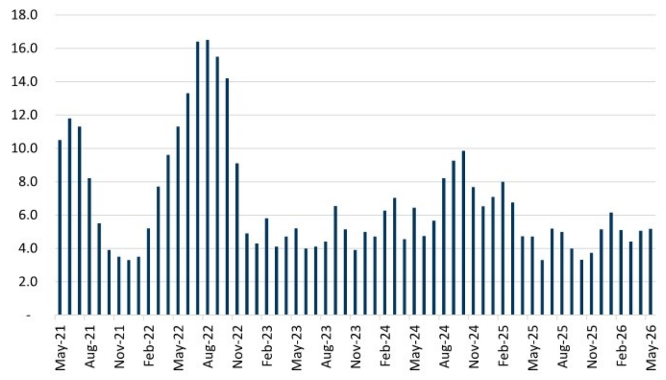
Source: Pacific Merchant Shipping Association

Exhibit 12: % of Containers Dwelling More than 5 Days



Source: Pacific Merchant Shipping Association

Exhibit 13: Rail Container Dwell Time, Days

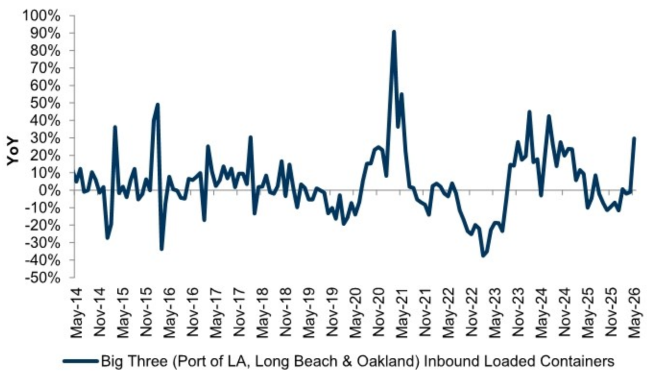


Source: Pacific Merchant Shipping Association

"Big Three" West Coast Ports' Inbound Loaded Containers

- **Total inbound containers for the Ports of LA, Long Beach, and Oakland +30% YoY in April. See our Big Three Note.**

Exhibit 14: West Coast Ports' Inbound Loaded Containers +30% YoY in April



Source: Port of Long Beach, Port of Los Angeles, Port of Oakland

Door to Door Shipping, China to US

- **It was taking an average of 47 days to ship (door to door) from China to the US in October, much closer to average pre-pandemic transit times when compared to peak congestion at 80+ days, and largely unchanged MoM vs 46 days in September. We assumed unchanged transit times in Nov-April for index updating purposes (given limited updates from data provider).**

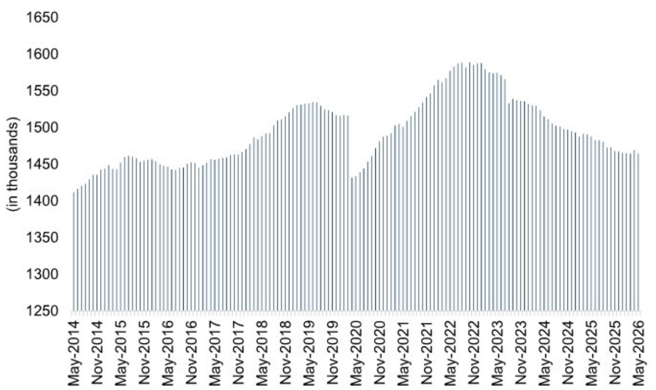
Exhibit 15: Door to Door Shipping Days, China to US

Source: Freightos

Trucking Employee Count

- **Truck transportation employee count was implied to be below pre-pandemic highs in May (-7.8% below pre-Covid); YoY growth has averaged -1.7% over the past six months.** May's employee count was -0.3% sequentially.

Exhibit 16: Total Truck Transportation Employee Count, Seasonally Adjusted



Source: US Bureau of Labor Statistics

LMI Capacity and Utilization

- **LMI Transportation Capacity Index**
  - Transportation capacity contracted in May at a slower rate compared to April given the 31.7 reading; the reading increased relative to April's 28.4 reading

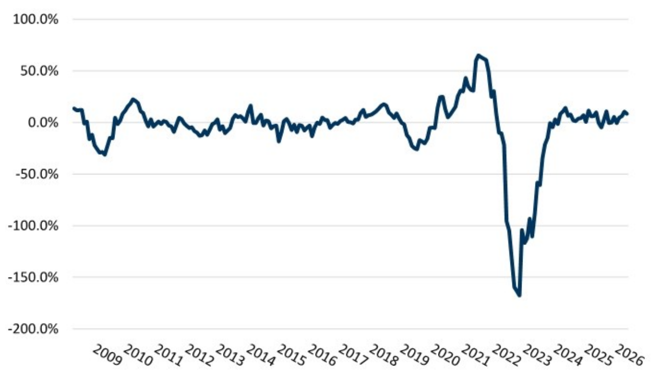
(implying a slower contraction of capacity).

- **LMI Warehouse Capacity Index**
  - Warehouse capacity was expanding in May given the 50.5 reading; this result was up from April (indicating more available overall warehouse capacity).
- **LMI Warehouse Utilization Index**
  - Warehouse utilization showed expansion in May at a slower pace compared to April (62.9 reading vs 64.4 in April).

PMI Supplier Delivery Times

- Delivery times were expanding in May (i.e., below 50 indicates expanding delivery times) given the 41.4 reading; the May supplier delivery PMI index was +8.4% YoY.

Exhibit 17: PMI: Manufacturing Suppliers' Delivery Times, YoY, Seasonally Adjusted



Source: IHS Markit

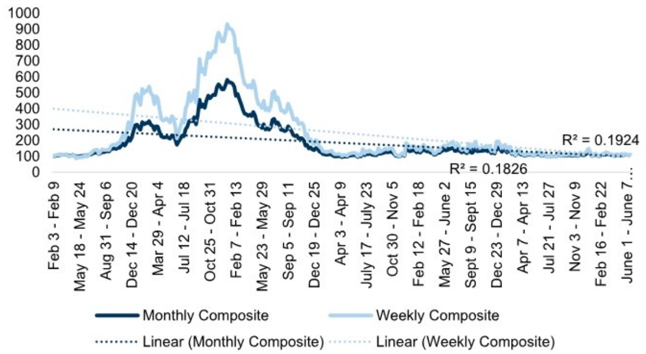
Appendix

Given the importance supply chain fluidity has on retailers, consumer goods companies, inflationary pricing, etc., we think this scale's importance is tied most directly to the pace at which supply chain congestion is on the mend. To this end, we look at a variety of variables that we think tie directly, or in some cases indirectly, to overall congestion; including ships at anchor, days to deliver, various dwell times, intermodal volume and velocity statistics amongst others. Aggregating this data, we create the Supply Chain Congestion Scale – an attempt to quantify the balance between supply chains being "Fully Bottlenecked" and "Fully Open," relative to the pre-pandemic benchmark we chose as Feb 3rd, 2020. Basically, how fluid is the overall transport logistics network.

To determine the position of the Legacy Congestion Scale (1-10), we calculate growth or decline in each category (monthly and weekly variables) relative to pre-congestion levels, overweighting certain categories we deem most directly tied to supply chain bottlenecks (i.e., ships anchored off the ports of LA and Long Beach, shipping container and chassis street dwell times, door-to-door shipping days from China to US), and scale it based on our Composite Scale (Exhibit 19).

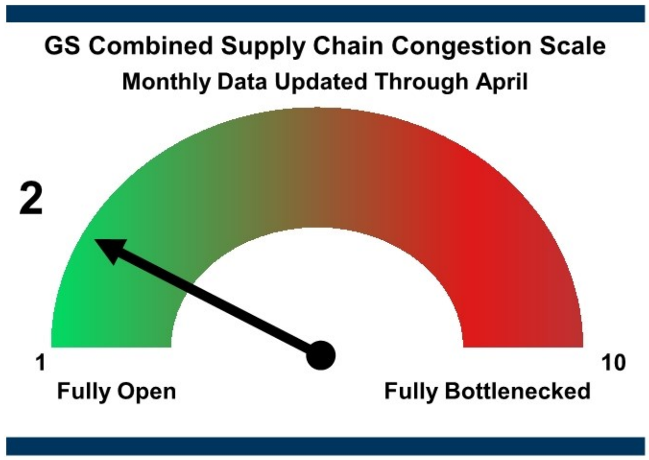
We publish the weekly scale on Monday PM/Tuesday AM to relay leading edge data that will inform the roughly one month lagged composite (i.e., we will update the weekly scale every week and the legacy scale will be updated monthly). Looking at the two charts, it is generally clear that the weekly composite should have good predictive ability as to the eventual direction of the monthly composite (which includes more encompassing variables than pure weekly ones to provide strong confirmation on congestion direction), and this is confirmed by the similar R^2 values seen in Exhibit 18.

Exhibit 18: The weekly composite index (light blue) leads the monthly (dark blue); expect future monthly updates to confirm recent weekly trends



Source: Goldman Sachs Global Investment Research

Exhibit 19: Our combined scale averaged '108' in May, indicating a bottleneck score of '2' but close to '1' when looking at all metrics (weekly and monthly combined)
Weekly + Monthly Combined Congestion Scale*



*We rescaled our index on 3/28/2022 to account for higher-than-anticipated peak bottleneck levels

Source: Goldman Sachs Global Investment Research

Exhibit 20: GS Legacy Supply Chain Congestion Scale (incorporates monthly and weekly data)

Composite Scale	Fluidity Scale
<101	1
101-153	2
154-206	3
207-259	4
260-313	5
314-366	6
367-419	7
420-472	8
473-526	9
>526	10

Source: Goldman Sachs Global Investment Research

Exhibit 21: GS Weekly Supply Chain Congestion Scale (High-Frequency Data)

Composite Scale	Fluidity Scale
<101	1
101-192	2
193-284	3
285-376	4
377-468	5
469-560	6
561-652	7
653-745	8
746-837	9
>838	10

Source: Goldman Sachs Global Investment Research

Glossary:

1. **West Coast container ship backlog:** tracks the number of container ships waiting to dock at the ports of LA and Long Beach; we add the number of ships anchored within 40 miles of the coast to the number of ships slow-steaming to port (AKA: ships loitering farther offshore) to get a better depiction of the true container ship backlog.

2. **East Coast container ship backlog:** estimates the number of container ships waiting to dock at the East and Gulf Coast ports in the US; based on satellite image data technology (Eikon), we add the number of container ships sitting for at least three days within 140 miles of a US port (accounts for ships to the right of 100 degrees of longitude to account for East Coast).
3. **Intermodal traffic:** tracks Y/Y intermodal carload volume growth for West Coast Class 1 Rails (BNSF and UNP). Accelerating volumes would indicate a more fluid supply chain for our purposes as more goods get moved through the system.
4. **Intermodal velocity:** tracks the average speed of the intermodal railcars (BNSF and UNP).
5. **Intermodal dwell:** tracks the average number of hours an intermodal railcar is idle (i.e., how long the car spends waiting at a terminal).
6. **Chassis dwell:** refers to the average time for chassis waiting on-terminal and on-street.
7. **Rail container dwell:** refers to the number of days a container waits to depart from a rail facility after being unloaded from an ocean carrier.
8. **Container weighted average dwell:** refers to the number of days a container stays at a marine terminal after being unloaded from an ocean carrier and taken off the premises by a truck.
9. **Ocean shipping rates:** refers to the average cost of shipping a container by ocean; we specifically track the cost of shipping from East Asia to the US West Coast via the FBX01 index from Freightos.
10. **Big Three West Coast Ports' inbound loaded containers:** refers to the Y/Y growth that the Big Three West Coast ports (LA, Long Beach, and Oakland) see in inbound loaded containers (i.e., how many more goods are moving through the port versus the same period last year).
11. **PMI manufacturing supplier delivery time index:** readings of 50 indicate no change in delivery times versus the prior month; readings above 50 indicate delivery times improved (faster supply chain) and readings below 50 indicated delivery times deteriorated sequentially (more delayed supply chain). The index is a result of IHS Markit's PMI business survey which asks purchasing managers, "Are your suppliers' delivery times slower, faster or unchanged on average than one month ago?"
12. **China-to-US door-to-door shipping transit time:** sourced from Freightos, this metric tracks the number of days it takes for a good to reach final destination (on average) once an order is placed and accepted (i.e., door-to-door shipping is not necessarily port-to-port as it also captures delivery times from port to final destination).

GS Data Works leverages alternative data sources and advanced analysis techniques to create unique data-driven insights across Global Investment Research. GS Data Works analysis (East Coast container ship backlog estimates) provided by Dan Duggan, Ph.D., Aditi Singh, and Parag Agrawal.

Disclosure Appendix

Reg AC

We, Jordan Alliger, Andrzej Tomczyk, CFA and Paul Stoddard, hereby certify that all of the views expressed in this report accurately reflect our personal views about the subject company or companies and its or their securities. We also certify that no part of our compensation was, is or will be, directly or indirectly, related to the specific recommendations or views expressed in this report.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

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GS Factor Profile

The Goldman Sachs Factor Profile provides investment context for a stock by comparing key attributes to the market (i.e. our universe of rated stocks) and its sector peers. The four key attributes depicted are: Growth, Financial Returns, Multiple (e.g. valuation) and Integrated (a composite of Growth, Financial Returns and Multiple). Growth, Financial Returns and Multiple are calculated by using normalized ranks for specific metrics for each stock. The normalized ranks for the metrics are then averaged and converted into percentiles for the relevant attribute. The precise calculation of each metric may vary depending on the fiscal year, industry and region, but the standard approach is as follows:

Growth is based on a stock's forward-looking sales growth, EBITDA growth and EPS growth (for financial stocks, only EPS and sales growth), with a higher percentile indicating a higher growth company. **Financial Returns** is based on a stock's forward-looking ROE, ROCE and CROCI (for financial stocks, only ROE), with a higher percentile indicating a company with higher financial returns. **Multiple** is based on a stock's forward-looking P/E, P/B, price/dividend (P/D), EV/EBITDA, EV/FCF and EV/Debt Adjusted Cash Flow (DACF) (for financial stocks, only P/E, P/B and P/D), with a higher percentile indicating a stock trading at a higher multiple. The **Integrated** percentile is calculated as the average of the Growth percentile, Financial Returns percentile and (100% - Multiple percentile).

Financial Returns and Multiple use the Goldman Sachs analyst forecasts at the fiscal year-end at least three quarters in the future. Growth uses inputs for the fiscal year at least seven quarters in the future compared with the year at least three quarters in the future (on a per-share basis for all metrics).

For a more detailed description of how we calculate the GS Factor Profile, please contact your GS representative.

M&A Rank

Across our global coverage, we examine stocks using an M&A framework, considering both qualitative factors and quantitative factors (which may vary across sectors and regions) to incorporate the potential that certain companies could be acquired. We then assign a M&A rank as a means of scoring companies under our rated coverage from 1 to 3, with 1 representing high (30%-50%) probability of the company becoming an acquisition target, 2 representing medium (15%-30%) probability and 3 representing low (0%-15%) probability. For companies ranked 1 or 2, in line with our standard departmental guidelines we incorporate an M&A component into our target price. M&A rank of 3 is considered immaterial and therefore does not factor into our price target, and may or may not be discussed in research.

Quantum

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Distribution of ratings/investment banking relationships

Goldman Sachs Investment Research global Equity coverage universe

	Rating Distribution			Investment Banking Relationships		
	Buy	Hold	Sell	Buy	Hold	Sell
Global	50%	34%	16%	65%	60%	45%

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